

Task 2 – Client Profitability and Spread Recommendation

We compute expected PnL per trade for each client at each closing horizon $\tau \in \{5,10,15,20,25,30\}$ using Corollary 1: $\mathbb{E}[\text{PnL}(\tau)] = \frac{1}{N} \sum \text{Side} \cdot V \cdot (M_\tau - TP)$. The expected aggregate PnL (Corollary 2) uses uniform closing weights: $\mathbb{E}[\text{Agg PnL}] = \frac{1}{N} \sum \text{Side} \cdot V \cdot (\bar{M} - TP)$ where $\bar{M}$ is the average of M_5 through M_{30} . All expectations are sample averages over each client's trades. A client is classified as **profitable** if expected aggregate PnL > 0, otherwise **costly**.

Results show Client B has expected aggregate PnL of +5.2 (profitable); Client C is marginally profitable at +0.3; Clients A (−19.4), D (−14.9), E (−19.2), and F (0.0) are costly. These classifications align with Task 1 adversity profiles: rising adversity (A, D) produces negative expected PnL; falling adversity (B) produces positive; flat near-50% adversity (C, F) yields near-zero PnL. Client E's falling adversity but large negative PnL indicates severe initial losses that dominate the average.

The minimum half-spread δ^* required for non-negative expected aggregate PnL is derived from quoting at $M_0 \pm \delta$. Substituting $TP = M_0 - \text{Side} \cdot \delta$ into the aggregate PnL formula gives $\mathbb{E}[\text{Agg PnL}] = \mathbb{E}[\text{Side} \cdot V \cdot (\bar{M} - M_0)] + \delta \cdot \mathbb{E}[V]$. Setting this ≥ 0 yields $\delta^* = \max(0, -\mathbb{E}[\text{Side} \cdot V \cdot (\bar{M} - M_0)]/\mathbb{E}[V])$. Computed values: Client A: 0.0450, B: 0.0000, C: 0.0000, D: 0.0382, E: 0.0478, F: 0.0000. Profitable or break-even clients require zero spread; costly clients need positive spreads proportional to their adverse selection cost.

The relationship to Task 1 adversity profiles is that adversity alone (percentage of losing trades) does not determine δ^* ; the magnitude of losses also matters. Clients with rising adversity (A, D) require $\delta^* > 0$ because losses become more frequent and larger over time. Client B, despite having 32–39% adversity, has positive expected PnL because losses are small and gains large – no spread needed. Client E has high but falling adversity, yet the losses are so severe that a large δ^* (0.0478) is still required. Thus δ^* captures both the frequency and severity of adverse selection, while the adversity profile only captures frequency.